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An Actual Method Proposal for Battery-Free Design

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Abstract

A successful device does not by itself validate the method that designed it. We report an actual model-authored method proposal in a frozen battery-free apparatus. The original nine-request proposal episode failed at the host action contract; a posthoc adapter recovered its rejected method text without another model decision. Fresh baseline and revised-method agents subsequently produced designs that both pass the complete computational service. Their first receptions occur at 46.9 and 5.6 milliseconds, respectively. These observations support distinct delivery strategies, but initial cache parity was not established and the baseline contradicts the method's claimed necessary capacitance band. The external decision rejects promotion and retains the experimental artifact. Neither run establishes state retention across reset, method advantage, heldout transfer, recursive self-improvement, or hardware qualification.

1 Question, apparatus, and intervention boundary

This finite study asks whether an actual proposed design method should replace the baseline method. Device changes D and design-method changes H have different validation scopes. Here D contains two device parameters: capacitance and the earliest allowed transmit time. H contains instructions for observing sources, forming energy and timing constraints, selecting a candidate, and checking it. A changed D requires its complete feedback response; a global change to H requires fresh design decisions from task entry. Changing the procedure that proposes methods, G, would require a separate lineage study. No such G comparison is reported.

The fixed public development task is `b-volatile-reset`. One event at $t = 0$ must deliver exactly one valid eight-byte application frame by 100 ms, without duplicates. Legal capacitance is 4–22 μF; the legal transmit floor is 0–45 ms. Supply, initial state, event, all other device fields, and the evaluator remain fixed. Harvest segments occur at 0–6 and 30–60 ms. The event latch is volatile. Passing means passing the complete frozen computational service, including independent energy and receiver checks and transient-voltage and deduplication checks. Physical calibration remains `UNKNOWN`; this is not a hardware or official benchmark result.

A registered device-response negative control distinguishes replay from valid reuse: the retained 10 μF / 45 ms design fails, whereas 10 μF / 0 ms passes on the same world. Reusing the former response for the changed input would give the wrong answer. The control read exact-input cached full responses and added no new execution. It was prepared by the host and is not an observation generated by the method proposer. The qualification-grid construction cost remains in the phase accounting.

2 Original failure and posthoc method recovery

The actual proposer used nine provider requests. Its live episode ended `INCOMPLETE_OR_STOPPED`: a host action was wrongly exposed and a device check payload consumed a method-proposal slot. No original method submission or adoption succeeded. A zero process exit code does not change that result. The host-action and budget-contract failure, trace, and all nine requests remain recorded.

A posthoc host adapter recovered an actual model-authored method from the retained rejected call. It added no paid proposal request or new model decision. The resulting artifact is eligible for experimental re-execution; it is neither a researcher-authored replacement nor a successful original proposal episode. Its method-text identity is bound as a canonical JSON string, with the hash encoding recorded in `phase2-binding.json`. This recovery is explicitly posthoc and does not repair the original prospective protocol.

The proposed H instructs the agent to acquire all listed sources, construct an energy and timing ledger, avoid losing the volatile latch before transmission, and check one selected design. It also makes device-specific exclusions: it presents 7.4–14 μF as a feasible band, rejects waiting across the dark gap or using a late transmit floor, and recommends 10 μF / 2 ms. These exclusions are scientific claims to test, rather than consequences of passing the method schema. Its declared falsifier includes failure of the selected design and an edge-design clause fixing a 2 ms floor and requiring selected-design failure. An outside-band success at a different floor does not, by itself, satisfy that exact conjunctive clause.

2.1 Fresh entries and comparison limits

The baseline H_0 and revised H' each ran from task entry with a distinct owner and context, zero initial decision messages, and a limit of two candidates, 24 provider requests, and 1,200 seconds. Each actually checked and submitted one candidate. The revised protocol appends the recovered method body verbatim to the common instruction. Its actual choice, 9 μF / 3.7 ms, differs from the method's 10 μF / 2 ms recommendation; this is a fresh decision, not a literal copy of that device recommendation.

The external registry reports matching task, source, information, model configuration, tool, and evaluator identities. Both use the provider-reported alias `deepseek-flash`; the weights are not independently pinned. Crucially, the physical exact-input cache evolved and its initial snapshot was not frozen equally for the two entries. The confirmation gate rejects with `PARITY_INITIAL_CACHE_SHA256`. Valid response reuse for a particular input does not establish equal starting caches. These runs therefore provide descriptive case outcomes, not a causal comparison of the methods.

3 Complete-service results and ordered state

Both designs pass the complete frozen computational service (Table 1). The external evaluator reads qualified complete responses whose input identities match the actual agent checks. The response receipts say `reused=true` and `origin=new-full-feedback-response`: reuse during external readback does not mean that the agent inherited an old decision trace, or that readback ran a new simulation. Both agent entries are fresh; cache construction is accounted separately.

Condition	C (μF)	TX floor (ms)	First RX (ms)	Frames	Service
H_0	22	45	46.9	1	PASS
H'	9	3.7	5.6	1	PASS

Table 1: One observed design per condition on the exposed fixed task. Physical qualification is UNKNOWN for both. These values do not estimate a success rate or a general method speedup.

The ordered-state audit resolves the reset interpretation. H_0 receives at 46.9 ms and has no reset before delivery or before the deadline. H' receives at 5.6 ms and resets at 12.1 ms, after delivery. Both have zero resets before first delivery, and both event latches remain volatile. The baseline avoids a pre-delivery reset; the revised design finishes delivery before reset. Neither successful run demonstrates that pending state survived a reset.

The frozen checker's legacy `event_preserved_after_reset` label is FAIL for H_0 and PASS for H'. Its implementation combines successful delivery with the occurrence of any reset; it does not test whether a still pending event crossed a reset. We retain those labels while interpreting the actual temporal order. In particular, the participant's broad "no brownout" description is unsupported over the full deadline: the post-delivery 12.1 ms reset remains in the audit. The checker was not changed.

3.1 Counterexample and retention decision

The passing 22 μF / 45 ms baseline is a counterexample to the proposed necessary 7.4–14 μF band and its rejection of waiting through the gap or using a late floor. It is not a counterexample to the separate claim that transmission must fit within one contiguous harvest segment: this baseline transmits during the second segment. Nor does it test the proposal's particular 22 μF / 2 ms edge case or the recommended 10 μF / 2 ms design. The revised design's success does not validate every rule in the method text.

The external decision is **reject promotion; retain the experimental artifact**. The default H is unchanged. This decision preserves the original host failure and posthoc boundary, the already successful baseline, the missing initial-cache parity, the cost of the proposal, and the over-restrictive method claims. The earlier reception under H' is an observed outcome; with one case per condition and unequal cache provenance, it does not establish a general or causal method advantage.

4 All requests, construction costs, and claim boundaries

Table 2 retains the four phase2 episodes, including the separate E5 consumer attempt and both original incomplete episodes. All 25 requests have settled usage; no usage is unresolved and no reservation remains unsettled. Amounts are token-based estimates at the existing pinned rates, not invoices. The upper calculation charges all prompt tokens at USD 0.30 per million and completion tokens at USD 1.20; the cache-aware calculation charges cache-hit prompt tokens at USD 0.006 and misses at USD 0.30 per million.

Episode	Requests	Upper (USD)	Cache-aware estimate (USD)
E5 attempt (incomplete)	6	0.0644115	0.035886444
H proposal (incomplete)	9	0.0424863	0.021487644
H_0 design	5	0.0393375	0.025263132
H' design	5	0.0304569	0.018602820
All phase2 episodes	25	0.1766922	0.101240040

Table 2: Actual usage readback. The E5 scientific interpretation is outside this method report; its cost remains in the whole-phase total.

The H proposal and the two design episodes together use 19 requests: USD 0.1122807 upper and USD 0.065353596 cache-aware estimate. Proposal plus H' uses 14 requests and USD 0.0729432 / 0.040090464, compared with H_0's five requests and USD 0.0393375 / 0.025263132. Although the H' design episode alone has a lower estimated cost, omitting the nine proposal requests would hide the method-development cost. No advantage under the same total budget is established. Codex assistance is unpriced and nonzero; total end-to-end cost remains UNKNOWN.

The entire finite phase records seven new local computational cases: four qualification-grid cases and three consumer/design cases. Native calls are zero. The ordered-state audit and this manuscript build add no model request, scientific execution, or native dispatch. The already recorded 22 MOCK tests verify the structural method API; they are reused here without a test or full verification rerun. They do not establish scientific method benefit.

4.1 Evidence and reproducibility

`phase2-binding.json` binds the original failed proposal, recovered method, fresh-entry protocols and actual tool traces, external service results, ordered-state audit, retention decision, and every cost ledger. It also binds this manuscript, the shared style and official English university logo, and the resulting PDF. Source hashes distinguish raw UTF-8 method text from its canonical JSON-string identity. `render_phase2.py build` validates those inputs and rebuilds only this separate paper in two TeX passes; it does not execute the scientific apparatus. Build and visual receipts are retained in `docs/research/phase2-20261020/methods/paper-final/`.

All observations concern an exposed public development task under a fixed ideal model. No heldout transfer, repeated-sampling estimate, independent method lineage, recursive self-improvement, hardware acceptance, or overall product acceptance is established. The prior A/R fixed-space NO_GO, strict FIFO failure, and NOT_ACCEPTED are preserved. The prior manuscript and PDF are preserved as separate artifacts.